

Project Safety and Reflection Report

Safety of GenAI

In developing MediaSleuth360, we carefully considered several key safety aspects of using generative AI:

- **Data Privacy:** We implemented protocols to ensure user-uploaded audio and video files are processed securely and are not retained longer than necessary. The files are removed after the task is completed.
- **Transparency:** The app indicates when content is AI-generated and its limitations. Users are informed by the message "NOTE: The responses are generated by AI and rely on the speech in the media; therefore, they can be inaccurate. Double verify important information."
- **Jailbreak Detection:** This was achieved by employing a proofreader AI assistant persona and verifying the prompts from the user before generating a response. It responds only if the query is related to content present in the uploaded file.
- **Output Accuracy:** To mitigate the risk of hallucinations while responding to user queries or inaccurate summaries, we tested with various large language models for high-quality summarization and used the ones with the best results. We also implemented fact-checking mechanisms to ensure the response is relevant to the media uploaded.
- **Error Handling:** We considered all the errors that can occur while getting a response from the AI model and place handling mechanism so the system doesn't crash.

Overall, to mitigate the risks, we:

- Conduct security and penetration testing
- Provide clear user guidelines on appropriate use
- Placed resistance mechanisms to prevent misuse

Lessons Learned and Reflections

Key lessons by developing MediaSleuth360:

- **Importance of prompt engineering:** Carefully crafting system prompts significantly improved the quality and reliability of AI-generated summaries and responses. It ensured apt generation and ignored irrelevant queries.
- **Balancing accuracy and creativity:** Finding the right temperature and top_p setting for the large language model to generate informative yet engaging content was crucial.
- **User feedback is invaluable:** Early user testing and peer review revealed unexpected responses and helped refine the UI/UX design.
- **Ethical considerations are ongoing:** We realized the need for continuous ethical assessment as new edge cases, jailbreak attempts, and potential misuse scenarios have emerged.
- **Interdisciplinary collaboration is critical:** Combining knowledge of NLP, UI/UX design, and LLMs led to a more robust and user-friendly application.

This project has deepened our understanding of GenAI's potential and limitations in media analysis. We've gained insights into the complexities of deploying AI responsibly and balancing innovation with ethical considerations.